

数据库实验报告

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实验题目： 获取模式

一、实验内容

REQUIREMENTS

- 1) Write a python program to get all table schemas of university database
- 2) generate all SQL DDL of above tables
- **3) generate all SQL of insertion for one of above tables**

二、实现方法

1、获取表结构

通过“show tables”获取所有表

依次用“describe table_name”获取表结构

```
11 # 获取所有表名
12 def get_tables(): 3用法
13     cursor.execute("show tables")
14     tables=[table[0] for table in cursor.fetchall()]
15     return tables
16
17 # 获取所有表结构
18 def get_table_schemas(): 1个用法
19     tables=get_tables()
20     table_schemas={}
21     for table in tables:
22         cursor.execute(f"describe {table}")
23         result=cursor.fetchall()
24         table_schemas[table]=result
25     return table_schemas
```

2、获取 DDL

通过“show tables”获取所有表

依次用“show create table table_name”获取表结构

```
28 # 获取所有DDL
29 def get_ddl(): 1个用法
30     tables=get_tables()
31     table_ddl={}
32     for table in tables:
33         cursor.execute(f"show create table {table}")
34         result=cursor.fetchone()
35         table_ddl[table]=result[1]
36     return table_ddl
```

3、生成 insert 语句

随机获取一张表并随机获取其一个数据元组

获取其所有属性名

利用字符串拼接生成 insert 语句

```
38 # 生成insert
39 def generate_insert(): 1个用法
40     table=random.choice(get_tables())
41     cursor.execute(f"""
42     select column_name
43     from information_schema.columns
44     where table_schema = 'dbscslab2018' and table_name = '{table}'
45     """)
46     cols=[col[0] for col in cursor.fetchall()]
47     cursor.execute(f"select * from {table}")
48     results=random.choice(cursor.fetchall())
49     values=[]
50     for result in results:
51         result=str(result)
52         if "Decimal" in result:
53             result=re.findall( pattern: r'((.*?))',result)
54         values.append(result)
55     insert=f"insert into {table} ("
56     for i in range(len(cols)):
57         insert += cols[i]
58         if i != len(cols)-1:
59             insert += ','
60     insert += ')\nvalues ('
61     for i in range(len(cols)):
62         insert += '\'+values[i]+'\'
63         if i != len(cols)-1:
64             insert += ','
65     insert += ');'
66     return insert
```

4、调用各函数并调整输出格式

```
70 print("(1) get all table schemas of university database")
71 table_schemas=get_table_schemas()
72 for schema in table_schemas:
73     print(f"{schema:<10}")
74     for col in table_schemas[schema]:
75         print("    "+str(col))
76     print("-----")
77
78 print("=====")
79 print("(2) generate all SQL DDL of above tables")
80 table_ddl=get_ddl()
81 for table in table_ddl:
82     print(table+":")
83     print(str(table_ddl[table]))
84     print("-----")
85
86
87 print("=====")
88 print("(3) generate all SQL of insertion for one of above tables")
89 table_insert=generate_insert()
90 print(table_insert)
```

三、结果展示（节选）

```
运行 test1 x
(1) get all table schemas of university database
advisor
('s_ID', 'varchar(5)', 'NO', 'PRI', None, '')
('i_ID', 'varchar(5)', 'YES', 'MUL', None, '')
-----
classroom
('building', 'varchar(15)', 'NO', 'PRI', None, '')
('room_number', 'varchar(7)', 'NO', 'PRI', None, '')
('capacity', 'decimal(4,0)', 'YES', '', None, '')
-----
course
('course_id', 'varchar(8)', 'NO', 'PRI', None, '')
('title', 'varchar(50)', 'YES', '', None, '')
('dept_name', 'varchar(20)', 'YES', 'MUL', None, '')
('credits', 'decimal(2,0)', 'YES', '', None, '')
-----
```

```
运行 test1 x
=====
(2) generate all SQL DDL of above tables
advisor:
CREATE TABLE `advisor` (
  `s_ID` varchar(5) COLLATE utf8mb3_bin NOT NULL,
  `i_ID` varchar(5) COLLATE utf8mb3_bin DEFAULT NULL,
  PRIMARY KEY (`s_ID`),
  KEY `i_ID` (`i_ID`),
  CONSTRAINT `advisor_ibfk_1` FOREIGN KEY (`i_ID`) REFERENCES `instructor` (`ID`) ON DELETE SET NULL,
  CONSTRAINT `advisor_ibfk_2` FOREIGN KEY (`s_ID`) REFERENCES `student` (`ID`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb3 COLLATE=utf8mb3_bin
-----
classroom:
CREATE TABLE `classroom` (
  `building` varchar(15) COLLATE utf8mb3_bin NOT NULL,
  `room_number` varchar(7) COLLATE utf8mb3_bin NOT NULL,
  `capacity` decimal(4,0) DEFAULT NULL,
  PRIMARY KEY (`building`,`room_number`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb3 COLLATE=utf8mb3_bin
-----
```

```
运行 test1 x
=====
(3) generate all SQL of insertion for one of above tables
insert into prereq (course_id,prereq_id)
values ('692','426');
```

四、结论

通过本次实验，加深了 python 与 mysql 的结合使用的能力，熟悉了查找数据库模式的方法，成功实现了实验需求。